

Phase 3: Project Design Deliverables

Document 03: Solution Architecture & Technical Data Models

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1. Architectural Vision & Foundational Principles

The **Automated Standard Laptop Procurement Solution** is architected as an event-driven, decoupled enterprise service model hosted on the ServiceNow Now Platform. By separating user engagement channels, process orchestration, relational data persistence, and external identity/asset integrations, the architecture provides extreme resilience, real-time auditability, and seamless upgradability across platform releases.

Foundational Architectural Principles

- Separation of Concerns (SoC):** Presentation logic in the Service Portal remains strictly independent of workflow execution, which executes asynchronously in the background.
- Strict Schema Conformance:** Relies 100% on out-of-the-box (OOTB) ServiceNow ITIL and ITAM data structures (`sc_request`, `sc_req_item`, `sc_task`, `alm_hardware`, `sysapproval_approver`). Zero custom tables are created, eliminating technical debt and upgrade friction.
- Elevated Execution Boundaries:** Workflows execute under the `System User` context, ensuring that cross-table writes to asset and governance records succeed without compromising granular end-user ACL restrictions.
- Relational Data Integrity:** Cross-table state synchronization guarantees that parent requests, child line items, operational tasks, and hardware configuration items maintain complete transactional alignment throughout their lifecycle.

2. High-Level 4-Tier Architecture

The system topology is organized into four distinct logical tiers, providing a clean separation between user interaction, workflow automation, database storage, and external enterprise systems.

```

graph TD
    subgraph Tier1 [Tier 1: Presentation & Channel Tier]
        SP[Service Portal / Employee Center /esc]
        MOB[ServiceNow Mobile Agent App]
        EML[Actionable Email & Push Notifications]
    end

    subgraph Tier2 [Tier 2: Business Logic & Orchestration Tier]
        FD[Flow Designer Engine]
        APPR[Approval & Escalation Engine]
        TASK_ENG[Sequential Task Provisioning Engine]
        ASSET_SYNC[HAM Asset Synchronization Engine]
        ERR_ENG[Global Flow Error Handler]
        SLA_ENG[Task SLA Engine contract_sla]
    end

    subgraph Tier3 [Tier 3: Platform Data & Persistence Tier]
        REQ[sc_request: Request Cart Header]
        RITM[sc_req_item: Line Item Context]
        TASK[sc_task: Operational Fulfillment Tasks]
        APPR_REC[sysapproval_approver: Approval Governance]
        ALM[alm_hardware: Hardware Asset Master]
        CI[cmdb_ci_computer: Configuration Item Master]
        USR[sys_user: Enterprise Identity Master]
        DEPT[cmn_department: Department Master]
        GRP[sys_user_group: Assignment Groups]
    end

    subgraph Tier4 [Tier 4: Enterprise Integration & Infrastructure Tier]
        ENTRA[Microsoft Entra ID / Active Directory SSO]
        VENDOR[Hardware Vendor Supply Chain API / EDI]
        MDM[Intune / Jamf Automated Provisioning Hooks]
        AUDIT[sys_audit & sys_flow_context Audit Log Engine]
        SMTP[Corporate SMTP / POP3 Mail Relay]
    end

    SP -->|HTTPS / REST API| RITM
    MOB -->|HTTPS / REST API| APPR_REC
    EML -->|Inbound Email Actions| APPR_REC

    RITM -->|Trigger: Record Created| FD
    FD --> APPR
    FD --> TASK_ENG
    FD --> ASSET_SYNC
    FD --> ERR_ENG
    FD --> SLA_ENG

    APPR --> APPR_REC
    TASK_ENG --> TASK
    ASSET_SYNC --> ALM
    ASSET_SYNC --> CI
    RITM --- REQ
    TASK --- RITM
    USR --- RITM
    USR --- APPR_REC
    DEPT --- USR
    GRP --- TASK

```

```
ENTRA -->|LDAP / SCIM Integration| USR
ALM -. ->|Inventory PO Sync| VENDOR
TASK -. ->|Staging Profiles| MDM
FD -->|Execution Logging| AUDIT
FD -->|Outbound Notifications| SMTP
```

Architectural Tier Descriptions

Tier 1: Presentation & Channel Tier

- **Service Portal / Employee Center (/esc):** Exposes the `Standard Laptop Order` catalog item to corporate employees. Provides dynamic field validation, responsive device layout, and real-time visual progress chevrons tracking request fulfillment.
- **ServiceNow Mobile Agent App:** Enables line managers and department heads to review laptop requisitions and execute instant approvals/rejections from mobile devices.
- **Actionable Notification Engine:** Transmits formatted HTML emails containing one-click mailto approval links, empowering approvers to authorize expenditures without logging into the platform UI.

Tier 2: Business Logic & Orchestration Tier

- **Flow Designer Engine:** High-performance, low-code event-driven engine executing the 17-step automated procurement workflow asynchronously in the background.
- **Approval & Escalation Subflows:** Manages manager presence validation, automated 48-hour reminders, and 5-day auto-escalation to department heads.
- **Sequential Task Provisioning Engine:** Programmatically generates and assigns `sc_task` records, enforcing prerequisite staging milestones prior to logistics dispatch.
- **HAM Asset Synchronization Engine:** Executes transactional database updates against `alm_hardware` and `cmdb_ci_computer` upon task completion.
- **Global Error Handler:** Intercepts unhandled runtime exceptions, generates Priority 2 incidents, annotates work notes, and notifies platform administrators.
- **Task SLA Engine (contract_sla):** Enforces 72-hour fulfillment commitments across technical staging and delivery logistics.

Tier 3: Platform Data & Persistence Tier

- Relational database tables providing persistent, ACID-compliant storage for requests (`sc_request`), line items (`sc_req_item`), tasks (`sc_task`), approvals (`sysapproval_approver`), hardware assets (`alm_hardware`), configuration items (`cmdb_ci_computer`), and foundation organizational data (`sys_user`, `cmn_department`, `sys_user_group`).

Tier 4: Enterprise Integration & Infrastructure Tier

- **Microsoft Entra ID / SCIM:** Authoritative source for employee identity, direct manager relationships, cost centers, and departmental assignments.
- **Audit & Logging Engine (sys_audit, sys_flow_context):** Immutable historical ledger capturing every parameter change, approver IP, timestamp, and workflow transition.
- **SMTP Gateway:** High-reliability enterprise mail relay managing transactional and actionable email delivery.

3. Component Integration Architecture

The component integration architecture illustrates the exact sequence of synchronous transactions, asynchronous event triggers, and data pill flows across system boundaries.

```
sequenceDiagram
    autonumber
    actor Employee as Employee / Requester
    participant Portal as Service Portal (/esc)
    participant CatalogEngine as Service Catalog Engine
    participant DB as Relational Database
    participant FlowEngine as Flow Designer Engine
    participant Approver as Line Manager
    participant Tech as Hardware Technician
    participant AssetDB as Hardware Asset DB (alm hardware)

    Employee->>Portal: Fills Laptop Request Form
    Portal->>CatalogEngine: Client Script & UI Policy Validation
    CatalogEngine->>DB: INSERT sc_request & sc_req_item
    DB-->>FlowEngine: Fire Trigger (sc_req_item Record Created)

    activate FlowEngine
    FlowEngine->>DB: UPDATE sc_req_item (Stage: waiting_for_approval)
    FlowEngine->>DB: INSERT sysapproval_approver
    FlowEngine->>Approver: Dispatch Actionable Approval Email

    Approver->>DB: UPDATE sysapproval_approver (State: Approved)
    DB-->>FlowEngine: Resume Flow (Approval State == Approved)
    FlowEngine->>DB: UPDATE sc_req_item (Stage: fulfillment, State: WIP)
    FlowEngine->>DB: INSERT sc_task (Task 1: Staging & Imaging)

    Tech->>DB: UPDATE sc_task (Link Serial #, State: Closed Complete)
    DB-->>FlowEngine: Resume Flow (Task 1 Closed Complete)
    FlowEngine->>DB: UPDATE sc_req_item (Stage: delivery)
    FlowEngine->>DB: INSERT sc_task (Task 2: Logistics & Dispatch)

    Tech->>DB: UPDATE sc_task (Tracking #, State: Closed Complete)
    DB-->>FlowEngine: Resume Flow (Task 2 Closed Complete)
    FlowEngine->>AssetDB: UPDATE alm_hardware (install_status: In Use, assigned_to: Requester)
    FlowEngine->>DB: UPDATE sc_req_item (Stage: complete, State: Closed Complete)
    FlowEngine->>DB: UPDATE sc_request (State: Closed Complete)
    FlowEngine->>Employee: Dispatch Delivery & Welcome Email
    deactivate FlowEngine
```

4. Entity Relationship Model (ERD) & Data Schema

The solution strictly adheres to the ServiceNow relational data model, ensuring full data normalization across foundation data, service requests, operational tasks, and asset records.

```
erDiagram
    sc_request ||--|{ sc_req_item : "contains (1 to N)"
    sc_req_item ||--|{ sc_task : "decomposes into (1 to N)"
    sc_req_item ||--o{ sysapproval_approver : "governed by (1 to N)"

    sys_user ||--o{ sc_request : "places request"
```

```
sys_user ||--o{ sc_req_item : "requested for"
sys_user ||--o{ sc_req_item : "opened by"
sys_user ||--o{ sysapproval_approver : "assigned approver"
sys_user ||--o| sys_user : "reports to manager"
```

```
cmn_department ||--o{ sys_user : "department members"
cmn_department ||--o| sys_user : "dept head"
```

```
sys_user_group ||--o{ sc_task : "assignment group"
sys_user ||--o{ sc_task : "assigned technician"
```

```
alm_hardware ||--o{ sc_task : "allocated in task"
alm_hardware ||--o| sys_user : "assigned to user"
```

```
sc_request {
    string sys_id PK
    string number UK
    reference requested_for FK
    reference opened_by FK
    string request_state
    string stage
    currency price
    datetime sys_created_on
}
```

```
sc_req_item {
    string sys_id PK
    string number UK
    reference request FK
    reference cat_item FK
    reference requested_for FK
    reference opened_by FK
    int state
    string stage
    string approval
    reference configuration_item FK
    datetime due_date
}
```

```
sc_task {
    string sys_id PK
    string number UK
    reference request_item FK
    reference assignment_group FK
    reference assigned_to FK
    string short_description
    string description
    int priority
    int state
    reference cmdb_ci FK
}
```

```
sys_user {
    string sys_id PK
    string user_name UK
    string name
    string email
    reference manager FK
}
```

```

        reference department FK
        boolean active
        reference cost_center FK
    }

    sys_user_group {
        string sys_id PK
        string name UK
        reference manager FK
        string email
        boolean active
    }

    cmn_department {
        string sys_id PK
        string name UK
        reference dept_head FK
        string id UK
    }

    alm_hardware {
        string sys_id PK
        string asset_tag UK
        string serial_number UK
        reference model FK
        reference ci FK
        int install_status
        string substatus
        reference assigned_to FK
    }

    sysapproval_approver {
        string sys_id PK
        reference sysapproval FK
        reference approver FK
        string state
        string comments
        datetime due_date
    }

```

Comprehensive Data Dictionary for Core Interacting Tables

1. Table: `sc_request` (Request Header Container)

Inherits from: `task` | Physical Table: `sc_request`

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Business Description
<code>sys_id</code>	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique global identifier.
<code>number</code>	Number	String (40)	Unique, Indexed	Auto	—	Human-readable identifier (e.g. REQ001020)

<code>requested_for</code>	Requested for	Reference	Mandatory, FK	Auto	<code>sys_user.sys_id</code>	Employee who receive the goods.
<code>opened_by</code>	Opened by	Reference	Mandatory, FK	<code>gs.getUserID()</code>	<code>sys_user.sys_id</code>	User who ph submitted the
<code>request_state</code>	Request state	Choice (String)	Mandatory	<code>requested</code>	—	Values: <code>sav</code> <code>requested</code> <code>in_proces</code> <code>closed_cc</code> <code>closed_in</code>
<code>stage</code>	Stage	Choice (String)	Mandatory	<code>requested</code>	—	Order cart s <code>requested</code> <code>approval</code> <code>fulfillme</code> <code>closed</code>
<code>price</code>	Price	Currency	Mandatory	<code>0.00</code>	—	Total price a from child li
<code>sys_created_on</code>	Created	GlideDateTime	Auto	Current Time	—	Exact datab creation tim

2. Table: `sc_req_item` (Requested Line Item)

Inherits from: `task` | Physical Table: `sc_req_item`

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target
<code>sys_id</code>	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—
<code>number</code>	Number	String (40)	Unique, Indexed	Auto	—
<code>request</code>	Request	Reference	Mandatory, FK	—	<code>sc_request.sys_</code>
<code>cat_item</code>	Item	Reference	Mandatory, FK	—	<code>sc_cat_item.sys.</code>
<code>requested_for</code>	Requested for	Reference	Mandatory, FK	—	<code>sys_user.sys_id</code>
<code>opened_by</code>	Opened by	Reference	Mandatory, FK	<code>gs.getUserID()</code>	<code>sys_user.sys_id</code>
<code>state</code>	State	Integer (Choice)	Mandatory	<code>1</code> (Open)	—

stage	Stage	Choice (String)	Mandatory	waiting_for_approval	—
approval	Approval	Choice (String)	Mandatory	requested	—
configuration_item	Configuration Item	Reference	Optional, FK	—	cmdb_ci_compute
due_date	Due date	GlideDateTime	Calculated	+72 Hours	—
work_notes	Work notes	Journal	Optional	—	—

3. Table: `sc_task` (Fulfillment Catalog Task)

Inherits from: `task` | **Physical Table:** `sc_task`

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Busin Descr
sys_id	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique task id
number	Number	String (40)	Unique, Indexed	Auto	—	Auto-g task nr (e.g., SCTAS
request_item	Request item	Reference	Mandatory, FK	—	sc_req_item.sys_id	Parent line ite
assignment_group	Assignment group	Reference	Mandatory, FK	—	sys_user_group.sys_id	Fulfillr (Hard Fulfi Group Logis
assigned_to	Assigned to	Reference	Optional, FK	—	sys_user.sys_id	Specif technic workin
short_description	Short description	String (160)	Mandatory	—	—	Concis popula Flow D
description	Description	String (4000)	Optional	—	—	Detaile hardwa specifi OS, bu address

priority	Priority	Integer (Choice)	Mandatory	3 (Moderate)	—	Values (Critical (High), Moderate (Low)).
state	State	Integer (Choice)	Mandatory	1 (Open)	—	Values (Open, Closed, Completed, Incomplete).
cmdb_ci	Configuration item	Reference	Conditional, FK	—	cmdb_ci_computer.sys_id	Mandatory closing (Staging, Imaging).

4. Table: sys_user (Enterprise User Identity)

Physical Table: sys_user

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Business Description
sys_id	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique global user identifier.
user_name	User ID	String (100)	Unique, Indexed	—	—	Corporate SSO username / login ID (e.g., jane.smith).
name	Name	String (160)	Mandatory	—	—	Full display name.
email	Email	String (100)	Indexed	—	—	Corporate email address used for notification delivery.
manager	Manager	Reference	Self-FK	—	sys_user.sys_id	Direct line manager responsible for approval sign-off.
department	Department	Reference	FK	—	cmn_department.sys_id	Corporate departmental alignment.
active	Active	Boolean	Mandatory	true	—	Employment status flag; inactive accounts bypass approvals.

cost_center	Cost center	Reference	FK	—	cmn_cost_center.sys_id	Financial accounting cost center charged for hardware.
-------------	-------------	-----------	----	---	------------------------	--

5. Table: sys_user_group (Assignment Groups)

Physical Table: sys_user_group

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Business Description
sys_id	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique group identifier.
name	Name	String (100)	Unique, Indexed	—	—	Group name (Hardware Fulfillment Group, IT Logistics).
manager	Manager	Reference	FK	—	sys_user.sys_id	Team manager responsible for group queue oversight.
email	Group email	String (100)	Optional	—	—	Shared distribution email for queue alerts.
active	Active	Boolean	Mandatory	true	—	Operational status of the group.

6. Table: cmn_department (Department Master)

Physical Table: cmn_department

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Business Description
sys_id	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique department identifier.
name	Name	String (100)	Unique, Indexed	—	—	Department title (e.g., Engineering, Finance).
dept_head	Department head	Reference	FK	—	sys_user.sys_id	Department Head executive used for approval escalations.
id	Department ID	String (40)	Unique	—	—	Corporate accounting department identifier.

7. Table: alm_hardware (Hardware Asset Master)

Inherits from: alm_asset | Physical Table: alm_hardware

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Business Description
-------------	-------	-----------	------------	---------	--------------------	----------------------

<code>sys_id</code>	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique as record identifier.
<code>asset_tag</code>	Asset tag	String (100)	Unique, Indexed	—	—	Physical label affixed to laptop.
<code>serial_number</code>	Serial number	String (100)	Unique, Indexed	—	—	OEM hardware serial number.
<code>model</code>	Model	Reference	Mandatory, FK	—	<code>cmdb_model.sys_id</code>	Reference to hardware specification model.
<code>ci</code>	Configuration item	Reference	FK	—	<code>cmdb_ci_computer.sys_id</code>	Linked Configuration record in the CMDB.
<code>install_status</code>	State	Integer (Choice)	Mandatory	<code>6</code> (In Stock)	—	Values: <code>1</code> (In Stock), <code>2</code> (In Transit), <code>7</code> (Retired).
<code>substatus</code>	Substate	Choice (String)	Conditional	<code>available</code>	—	Values when In Stock: <code>available</code> , <code>reserved</code> , <code>pending</code> .
<code>assigned_to</code>	Assigned to	Reference	FK	—	<code>sys_user.sys_id</code>	Employee responsible for possession of machine; on delivery.

8. Table: `sysapproval_approver` (Approval Governance Records)

Physical Table: `sysapproval_approver`

Column Name	Label	Data Type	Constraint	Default	Foreign Key Target	Business Description
<code>sys_id</code>	Sys ID	GUID (32 hex)	PK, Mandatory	Auto	—	Unique approval record identifier.
<code>sysapproval</code>	Approval for	Reference	Mandatory, FK	—	<code>task.sys_id</code> (<code>sc_req_item</code>)	The target line item requiring governance sign-off.
<code>approver</code>	Approver	Reference	Mandatory, FK	—	<code>sys_user.sys_id</code>	Designated authority responsible for authorizing request.

state	State	Choice (String)	Mandatory	requested	—	Values: not_requested, requested, approved, rejected.
comments	Approval comments	Journal	Optional	—	—	Formal justification comments recorded by approver.
due_date	Due date	GlideDateTime	Calculated	+5 Days	—	Deadline for escalation before rerouting to Dept Head.

5. State-Transition Architecture & Synchronization Rules

To maintain absolute data integrity across parent requests, child items, operational tasks, and asset inventory, the system enforces a strict **State Machine Architecture**.

5.1 Parent Request (sc_request.request_state) State Machine

```
stateDiagram-v2
    [*] --> requested: Cart Committed
    requested --> in_process: Child RITM Flow Instantiated
    in_process --> closed_complete: All Child RITMs Closed Complete
    in_process --> closed_incomplete: Child RITM Rejected or Incomplete
    in_process --> closed_cancelled: Requester Manually Aborts
    closed_complete --> [*]
    closed_incomplete --> [*]
    closed_cancelled --> [*]
```

5.2 Requested Item (sc_req_item) Dual State & Stage Engine

ServiceNow utilizes a dual-layer tracking mechanism: **stage** drives the customer-facing progress chevrons in the Service Portal, while **state** governs backend SLA calculation and task workflow states.

```
stateDiagram-v2
    [*] --> WaitingForApproval: Flow Triggered
    note right of WaitingForApproval
        stage = waiting_for_approval
        state = 1 (Open)
        approval = requested
    end note

    WaitingForApproval --> Fulfillment: Manager Approves
    note right of Fulfillment
        stage = fulfillment
        state = 2 (Work in Progress)
        approval = approved
```

```

    Task 1 Provisioned
end note

WaitingForApproval --> RequestCancelled: Manager Rejects
note right of RequestCancelled
    stage = request_cancelled
    state = 4 (Closed Incomplete)
    approval = rejected
    Flow Aborts
end note

Fulfillment --> Delivery: Task 1 (Staging) Closed Complete
note right of Delivery
    stage = delivery
    state = 2 (Work in Progress)
    Task 2 Provisioned
end note

Delivery --> Complete: Task 2 (Logistics) Closed Complete
note right of Complete
    stage = complete
    state = 3 (Closed Complete)
    Asset install_status = In Use
end note

RequestCancelled --> [*]
Complete --> [*]

```

5.3 Catalog Task (sc_task.state) State Machine

```

stateDiagram-v2
    [*] --> Open: Flow Creates Task
    note right of Open
        state = 1 (Open)
        Assigned to Fulfillment Group
    end note

    Open --> WIP: Technician Accepts Task
    note right of WIP
        state = 2 (Work in Progress)
        assigned_to Populated
    end note

    WIP --> ClosedComplete: Work Finished Successfully
    note right of ClosedComplete
        state = 3 (Closed Complete)
        Mandatory Asset Fields Validated
    end note

    WIP --> ClosedIncomplete: Hardware Out of Stock / Unreachable
    note right of ClosedIncomplete
        state = 4 (Closed Incomplete)
        Triggers Procurement Purchasing Subflow
    end note

    ClosedComplete --> [*]

```

```
ClosedIncomplete --> [*]
```

5.4 Hardware Asset (alm_hardware.install_status & substatus) Lifecycle

```
stateDiagram-v2
    [*] --> InStockAvailable: Physical Inventory Received
    note right of InStockAvailable
        install_status = 6 (In Stock)
        substatus = available
    end note

    InStockAvailable --> InStockReserved: Manager Approves Order
    note right of InStockReserved
        install_status = 6 (In Stock)
        substatus = reserved
    end note

    InStockReserved --> InTransit: Task 1 Closes & Device Dispatched
    note right of InTransit
        install_status = 2 (In Transit)
        substatus = ''
    end note

    InTransit --> InUse: Task 2 Closes & Delivery Confirmed
    note right of InUse
        install_status = 1 (In Use)
        assigned_to = requested_for
    end note

    InUse --> [*]
```

5.5 Unified Cross-Table State Synchronization Matrix

The following matrix defines the exact relational state alignment across all seven core entity attributes at each operational milestone:

#	Business Milestone	sc_request.request_state	sc_req_item.stage	sc_req_item.state	sc_req_item.
1	Order Submitted	requested	waiting_for_approval	1 (Open)	requested
2	Manager Approves	in_process	fulfillment	2 (Work in Prog)	approved
3	Manager Rejects	closed_incomplete	request_cancelled	4 (Closed Incomp)	rejected
4	Tech Begins Staging	in_process	fulfillment	2 (Work in Prog)	approved

5	Staging Completed	in_process	delivery	2 (Work in Prog)	approved
6	Tech Ships Device	in_process	delivery	2 (Work in Prog)	approved
7	Verified Delivery	closed_complete	complete	3 (Closed Comp)	approved

Synchronization Enforcement Mechanisms

- **Flow Designer Actions:** Primary orchestrator driving all progressive forward transitions from Milestone 1 through 7.
- **ServiceNow Business Rules (Fail-Safe):** An out-of-the-box business rule (`sc_req_item` closure checks) validates parent `sc_request` state upon child closure to prevent orphaned open request containers.
- **Race Condition Mitigation:** Database row locks (`GlideRecord.setWorkflow(true)`) and sequential `Wait for Condition` action blocks ensure that Task 2 cannot be created until the database commit for Task 1 closure has completed.

6. Security, Governance & Role Architecture

Platform security enforces the principle of least privilege, strict segregation of duties, and regulatory compliance across all interacting personas.

```
graph TD
  ADMIN[admin: Full Platform Administration]
  CAT_ADMIN[catalog_admin: Catalog Items & Variable Policies]
  APPR_ADMIN[approval_admin: Approval Delegation & Reassignment]
  ITIL[itil: Fulfillment Specialist & Asset Handler]
  SNC_INT[snc_internal: Corporate Employee / Requester]

  ADMIN --> CAT_ADMIN
  ADMIN --> APPR_ADMIN
  CAT_ADMIN --> ITIL
  APPR_ADMIN --> ITIL
  ITIL --> SNC_INT
```

6.1 Role Definitions & Permission Scope

Role Identifier	Security Scope	Key Capabilities & Functional Privileges
snc_internal	Employee Center, <code>sc_cat_item</code> , self <code>sc_req_item</code>	Standard corporate employee. Can submit catalog items, view own submitted requests (<code>requested_for == gs.getUserID()</code>), post public comments. Cannot view internal technician work notes, task queues, or asset tables.
itil	<code>sc_task</code> <code>sc_req_item</code> , <code>alm_hardware</code> , <code>cmdb_ci</code>	Operational fulfiller role. Can view team task queues (<code>Hardware Fulfillment Group: IT Logistics</code>), update task states, enter work notes, scan serial numbers, and perform asset allocations.

approval_admin	sysapproval_approver	Governance oversight role. Can reassign stalled approvals, manage approval delegation during extended manager absences, and review enterprise approval audit logs.
catalog_admin	sc_cat_item, item_option_new, catalog_ui_policy	Service catalog management. Can configure catalog item specifications, variable sets, dynamic UI policies, and client scripts. Cannot alter core platform ACLs or database dictionaries.
admin	Instance-Wide	Full administrative control. Configures global Flow Designer engine properties, system dictionaries, access control lists (ACLs), and platform integration endpoints.

6.2 Access Control List (ACL) Matrix

The table below defines the security boundaries across core procurement tables for Create, Read, Write, and Delete operations:

Table Name	Operation	Role Required	Enforcement Condition / Script Safeguard
sc_request	create	snc_internal	Created automatically via Service Catalog cart checkout.
sc_request	read	snc_internal	`opened_by == gs.getUserID()
sc_request	write	itil	User possesses itil role and parent record is not in a closed state.
sc_request	delete	admin	Blocked in production, requests must be closed or cancelled, never deleted.
sc_req_item	create	snc_internal	Created automatically via Service Catalog submission.
sc_req_item	read	snc_internal	opened_by == gs.getUserID()
sc_req_item	write	itil	System flow executes with elevated System User rights; manual fulfiller writes restricted to work notes.
sc_task	create	system / admin	Programmatic creation restricted strictly to Flow Designer engine; manual user creation disabled.
sc_task	read	itil	User has itil role and is a member of assignment_group or holds admin .

sc_task	write	itil	Technician must be an active member of <code>current.assignment_group</code> to edit or close task.	
alm_hardware	read	itil	Read access permitted to ITIL users for asset tagging and verification.	
alm_hardware	write	asset / itil	Enforces mandatory audit logging when <code>install_status</code> or <code>assigned_to</code> fields are modified.	
sysapproval_approver	read	snc_internal	<code>`approver == gs.getUserID()</code>	<code>sysapproval.opened_by == gs.getUserID()</code>
sysapproval_approver	write	snc_internal	<code>`approver == gs.getUserID()</code>	<code>gs.hasRole('approval_admin')</code> (Only assigned approver can approve).

6.3 Data Privacy, PII Isolation & Audit Immutability

1. PII Isolation for Remote Deliveries:

- Variables containing private home delivery details (`shipping_address`, `contact_phone`) are protected using **Variable Read Roles**. General employees browsing portal requests cannot view private residential addresses; visibility is restricted strictly to the submitting employee, their line manager, and members of **IT Logistics & Deskside Support**.

1. Read-Only Data Lock Post-Approval:

- A Catalog UI Policy automatically locks all hardware option choices as Read-Only once the request transitions out of the initial `Open` state. This prevents tampering or modifying specifications while orders are actively being fulfilled.

1. Immutable Forensic Audit Ledger:

- High-fidelity auditing is active on `sc_req_item`, `sc_task`, and `sysapproval_approver` via the platform audit engine (`sys_audit`). All field updates, approval decisions, user IPs, and workflow engine transitions are permanently archived, guaranteeing compliance with enterprise regulatory standards (SOX, ISO 27001).

7. Document Sign-off & Revision History

Version	Date	Author	Role	Description of Change
1.0	2026-09-30	worker_phase3	Phase 3 Design Lead	Authoritative Deliverable: 4-Tier Architecture Diagram, Component Integration Model, Full 8-Table ERD & Data Dictionary, State-Transition Architecture, and RBAC Security Matrix.